

Simple Interest

① Principal:- The money borrowed or lent out for a certain period is called the principal or the sum.

② Interest:- Extra money paid for using other's money is called interest.

③ Simple Interest (S.I.): If the interest on a sum borrowed for a certain period is reckoned uniformly, then it is called simple interest.

Let principle = P , Rate = $R\%$ per annum (p.a.)
and time = T Years Then,

$$i) \quad S.I. = \left(\frac{P \times R \times T}{100} \right)$$

Ex:-1 Find the simple interest on Rs. 68,000 at $18\frac{2}{3}\%$ per annum for 9 months

Solⁿ $P = 68,000$, $R = \frac{56}{3}\% \text{ p.a.}$ & $T = \frac{9}{12}$ years

$$\therefore S.I. = \left(\frac{P \times R \times T}{100} \right) = \text{Rs.} \left(68,000 \times \frac{56}{3} \times \frac{3}{4} \times \frac{1}{100} \right) = 85,000$$

Ex:-2 Find the simple interest on Rs. 3000 at $6\frac{1}{4}\%$ per annum for the period from 4th Feb. 2005 to 18th April, 2005.

Solⁿ Time = $(24 + 31 + 18)$ days = 73 days. = $\frac{73}{365}$ year

$T = \frac{1}{5}$ year, $P = 3000$ & $R = 6\frac{1}{4}\% \text{ p.a.} = \frac{25}{4}\% \text{ p.a.}$

S.I. = Rs. $\left(3000 \times \frac{25}{4} \times \frac{1}{5} \times \frac{1}{100} \right) = \text{Rs. } 37.50$

Remark:- The day on which money is deposited is not counted while the day on which money is withdrawn is counted.

Ex:-3 A sum at simple interest at $13\frac{1}{2}\%$ per annum amounts to Rs. 2502.50 after 4 years. Find the sum.

Solⁿ Let the sum be x

Then S.I. = Rs. $\left(x + \frac{27}{7} \times 4 \times \frac{1}{100} \right) = \frac{27x}{50}$

$\therefore \text{Amount} = \text{Rs.} \left(x + \frac{27x}{50} \right) = \frac{77x}{50} \text{ Rs.}$

$\therefore \frac{77x}{50} = 2502.50 \Rightarrow x = \frac{2502.50 \times 50}{77} = 1625$

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Ex-4 A sum of Rs. 800 amounts to Rs. 920 in 3 years at simple interest. If the interest rate is increased by 3%, it would amount to how much?

Soln
 $S.I. = Rs. (920 - 800) = Rs. 120.$
 $P = 800, T = 3 \text{ Years}$

$$\therefore R = \left(\frac{100 \times 120}{800 \times 3} \right) \% = 5\%$$

$$\text{New rate} = (5 + 3) \% = 8\%.$$

$$\text{New S.I.} = Rs. \left(\frac{800 \times 8 \times 3}{100} \right) = Rs. 192.$$

$$\text{New Amount} = Rs. (800 + 192) = Rs. 992.$$

Ex-5 Adam borrowed some money at the rate of 6% p.a. for the first two years, at the rate of 9% p.a. for the next three years and at the rate of 14% p.a. for the period beyond five years. If he pays a total interest of Rs. 11400 at the end of nine years, how much money did he borrow?

Soln Let the sum borrowed be x . Then,

$$\left(\frac{x \times 6 \times 2}{100} \right) + \left(\frac{x \times 9 \times 3}{100} \right) + \left(\frac{x \times 14 \times 4}{100} \right) = 11400$$

$$\Rightarrow \frac{3x}{25} + \frac{27x}{100} + \frac{14x}{25} = 11400 \Rightarrow \frac{95x}{100} = 11400$$

$$\Rightarrow x = \frac{11400 \times 100}{95} = 12,000.$$

Hence, sum borrowed = Rs. 12,000.

* Ex:-6 A certain sum of money amounts to Rs. 1008 in 2 years and to Rs. 1164 in $3\frac{1}{2}$ years. Find the sum and the rate of interest.

Soln S.I. for $1\frac{1}{2}$ years = Rs. $(1164 - 1008) = \text{Rs. } 156$

$$\text{S.I. for 2 years.} = \text{Rs. } \left(156 \times \frac{4}{3} \times 2 \right) = \text{Rs. } 408.$$

$$\text{Principle} = \text{Rs. } (1008 - 408) = \text{Rs. } 600.$$

Now, $P = 600$, $T = 2$, and $\text{S.I.} = 408$.

$$\text{Rate} = \frac{100 \times 408}{600 \times 2} \% = 13\%.$$

Ex: 7 At what rate percent per annum will a sum of money double in 16 years?

Solⁿ Let principle = P. Then S.I. = P

∴ T = 16 years.

$$\text{Rate} = \left(\frac{100 \times P}{P \times 16} \right) \% = 6\frac{1}{4}\% \text{ p.a.}$$

Ex: 8 The simple interest on a sum of money is $\frac{4}{9}$ of the principle. Find the rate percent and time, if both are numerically equal.

Solⁿ Let sum = Rs. x Then S.I. = Rs. $\frac{4x}{9}$

Let rate = R% ∴ time = R years

$$\text{Then } \left(\frac{x \times R \times R}{100} \right) = \frac{4x}{9} \quad \text{or } R^2 = \frac{400}{9}$$

$$\text{or } R = \frac{20}{3} = 6\frac{2}{3}$$

∴ Rate = $6\frac{2}{3}\%$ And time $6\frac{2}{3}$ years
= 6 yrs. 8 months

Ex:-9 The simple interest on a certain sum of money for $2\frac{1}{2}$ years at 12% per annum is Rs. 40 less than the simple interest on the same sum for $3\frac{1}{2}$ years at 10% per annum. Find the sum.

Solⁿ Let the sum be Rs. x .

$$\text{Then } \left(\frac{x \times 10 \times 7}{100 \times 2} \right) - \left(\frac{x \times 12 \times 5}{100 \times 2} \right) = 40$$

$$\Rightarrow \frac{7x}{20} - \frac{3x}{10} = 40 \Rightarrow x = 40 \times 20 = 800.$$

Hence the sum is Rs. 800.

Ex:-10 A sum was put at simple interest at a certain rate for 3 years. Had it been put at 2% higher rate, it would have fetched Rs. 360 more. Find the sum.

Solⁿ Let sum = P and original rate = R

$$\text{Then } \left[\frac{P \times (R+2) \times 3}{100} \right] - \left[\frac{P \times R \times 3}{100} \right] = 360$$

$$\Rightarrow 3PR + 6P - 3PR = 36000$$

$$\Rightarrow 6P = 36000 \Rightarrow P = 6000.$$

Hence, sum = Rs. 6000.

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